

VAYUNET

Understanding the Problem & Our Solution

A simple, plain-language guide to SIH26077
Hyper-Local Severe-Weather Nowcasting Engine

This document explains what the problem actually is, and what we are building to solve it — written for someone who is technically educated, but not a meteorologist or AI researcher.

1. What Is the Problem, Really?

What does “nowcasting” mean?

Most people are familiar with a weather forecast: a prediction of what the weather will be like over the next few days. “Nowcasting” is different — it means predicting what the weather will do in the very near future, typically the next few minutes to a few hours. Think of it as the difference between “will it rain this week” and “is it about to start raining heavily in my neighborhood right now.”

The forecasting family, from broad to precise

Type of Prediction	How Far Ahead	How Precise a Location
Weather forecasting	Days to a week ahead	State / large region
Short-range forecasting	1–3 days ahead	District-level
Nowcasting	0–6 hours ahead	Sub-district / city
Hyper-local nowcasting	0–6 hours ahead	Neighborhood / valley-level

VAYUNET is being built for the last row: hyper-local nowcasting. The goal isn’t just to say “it might rain today” — it’s to say “in the next 30 to 120 minutes, this specific valley or town is at high risk of a dangerous storm.”

A regular weather forecast might tell you that a district has a chance of thunderstorms today. VAYUNET aims to narrow that down — telling you which smaller areas within that district are likely to experience genuinely dangerous weather in the next couple of hours, with enough warning time to actually do something about it.

Why existing forecasts aren't enough for this

- Traditional weather models (called Numerical Weather Prediction, or NWP) work at a coarse resolution — often 25 kilometers or more per grid cell. A single grid cell can cover an entire district, so the forecast can't tell you which specific town within that district will actually get hit.
- These models also take time to run — often several hours — which eats into the warning time people actually have to react.
- Severe local storms like thunderstorms and cloudbursts develop and intensify very quickly, sometimes in under an hour. A forecast issued this morning may already be outdated by the time the storm forms.

Why India needs this specifically

- India's geography — the Himalayan foothills, the Western Ghats, and the monsoon system — creates conditions where local terrain strongly influences where storms form and intensify.
- Events like the Kedarnath (2013) and Chamoli (2021) disasters showed how a highly localized, rapidly developing weather event can cause massive loss of life when the warning wasn't precise or fast enough.
- A large population lives in hilly, flood-prone, or otherwise vulnerable regions where even a few hours of accurate, specific warning can mean the difference between evacuation and disaster.

Why thunderstorms and cloudbursts are hard to predict

- They are caused by fast-moving, small-scale atmospheric processes — rising warm, moist air interacting with unstable atmospheric layers — that can develop in under an hour, much faster than large-scale weather systems.
- Terrain plays a huge role (mountains force air upward, intensifying rainfall in specific valleys) but terrain effects are very local and hard to capture in coarse-resolution models.
- The events we care about most (extreme cloudbursts, flash floods) are rare by definition, which makes it statistically harder to train a reliable prediction system — there just isn't as much example data for the worst-case scenarios compared to normal weather.

Why a 0–6 hour prediction window is valuable

Zero to six hours is short enough to reflect what's actually happening in the atmosphere right now (rather than a days-old forecast), but long enough to matter operationally — enough time to alert a village, close a road, or move people away from a riverbed before the water rises. This is the window where hyper-local nowcasting adds real, actionable value that a standard forecast cannot.

2. How VAYUNET Solves This, Step by Step

VAYUNET works like an assembly line: raw information comes in one end, and a clear, actionable warning comes out the other. Here is the full pipeline in plain terms:

Data

What goes in: Satellite images, atmospheric measurements, and terrain maps

What happens: Raw information about the current state of the sky and the ground

Why it's necessary: The system needs to know what's happening in the atmosphere and on the ground before it can predict anything.

What comes out: A stack of images and numbers, aligned to the same map grid and time.

Preprocessing

What goes in: The raw data from the step above

What happens: Cleaning, aligning, and resizing everything onto one consistent map grid and time schedule

Why it's necessary: Satellite data, atmospheric data, and terrain data all come in different formats, resolutions, and update schedules — they have to be lined up before they can be compared or combined.

What comes out: A clean, uniform “stack” of data covering the same area and moments in time.

Feature Extraction

What goes in: The cleaned, aligned data

What happens: Calculating specific meteorological quantities known to signal severe weather (like atmospheric moisture buildup or instability)

Why it's necessary: Raw satellite brightness values aren't directly meaningful — they need to be converted into the physical quantities meteorologists actually use to judge storm risk.

What comes out: A set of numbers per location representing moisture, instability, and temperature-related risk signals.

AI Models

What goes in: The extracted features, as a time sequence

What happens: A trained neural network looks at how these features have been changing over the last few hours and learns to recognize the early warning patterns of a developing storm

Why it's necessary: This is the core prediction step — humans can't manually track thousands of grid points changing every 15–30 minutes, but a trained model can.

What comes out: A raw prediction of what the atmosphere will look like a few hours from now.

Multi-Source Fusion

What goes in: Predictions and features from satellite, atmosphere, and terrain sources

What happens: Combining all these different signals into one unified risk assessment, factoring in how the local terrain (valleys, slopes) would channel any resulting rainfall

Why it's necessary: No single data source tells the whole story — satellite alone can't see terrain effects, and terrain alone can't see incoming moisture. Combining them gives a fuller picture.

What comes out: A combined atmospheric-plus-terrain risk signal for each location.

Nowcasting

What goes in: The combined risk signal

What happens: Projecting this signal forward in time, typically in 30-minute steps, out to a 2–6 hour horizon

Why it's necessary: This is what makes it “nowcasting” rather than just “analysis of right now” — it explicitly looks ahead.

What comes out: A short-term forecast timeline of risk for the next several hours.

Risk Estimation

What goes in: The forecast timeline

What happens: Converting the raw numbers into an easy-to-understand risk category, like Low, Moderate, High, or Severe

Why it's necessary: A probability number (like “62% chance”) is hard for a non-expert to act on quickly — a risk category is immediately understandable.

What comes out: A risk level for each location and time step.

Alert

What goes in: The risk levels

What happens: Automatically generating a short, clear warning message when risk crosses a dangerous threshold

Why it's necessary: The whole point of the system is to help people act in time — a warning has to actually reach and make sense to the people who need it.

What comes out: A plain-language alert message (potentially in a local language), ready to be sent out.

Visualization

What goes in: Risk levels, alerts, and underlying data

What happens: Displaying everything on an interactive map dashboard that officials or the public can view

Why it's necessary: Numbers and alerts are far easier to understand, trust, and act on when shown visually on a map than as a table of numbers.

What comes out: A live, interactive risk map that updates as new predictions come in.

3. Where the Information Comes From

Satellite data

India's own weather satellites — INSAT-3D, INSAT-3DR, and the newer INSAT-3DS — continuously watch the atmosphere from space and are the main public source of this kind of data for India, distributed through ISRO's MOSDAC portal. These satellites capture several types of images: visible-light images (like a regular photo, only useful in daylight), infrared images (which reveal cloud-top temperature, useful day or night), and water-vapour images (which show how much moisture is present in the atmosphere, a key early-warning signal before a storm forms).

MOSDAC offers both raw satellite data and already-processed “derived products” (like cloud-top temperature or rainfall estimates, already calculated for you). VAYUNET should preferably use the derived products where possible — they save significant processing effort and have already been quality-checked by ISRO's own scientists, though the raw channels remain available if a specific custom calculation is ever needed.

Atmospheric / reanalysis data

Satellite images alone don't tell you about temperature, pressure, and humidity at different heights in the atmosphere — for that, VAYUNET uses “reanalysis” datasets, which are essentially detailed, gridded historical weather records built by combining many observation sources with physics-based models. The main option for India is IMDAA (built jointly by India's Ministry of Earth Sciences and the UK Met Office), with the global ERA5 dataset from Europe's weather agency as a backup. From these, VAYUNET calculates specific storm-risk indicators that meteorologists have relied on for decades — for example, a measure of how much “fuel” is available for a thunderstorm to grow (called CAPE), and a measure of atmospheric moisture buildup.

Terrain data

A digital elevation map — essentially a very detailed 3D map of the ground's height everywhere — tells VAYUNET how rainfall will actually flow once it hits the ground. Mountains force moist air upward, which increases rainfall on the windward side (this is called orographic rainfall) and channels water into specific valleys. India's own Cartosat-1 DEM (freely available via ISRO's Bhuvan portal) along with international options like SRTM and Copernicus DEM all provide this at roughly 30-meter detail, which is precise enough to identify which specific valleys are most at risk of flooding.

Radar

Weather radar (used by IMD's Doppler Weather Radar network) is extremely valuable for short-term rainfall tracking because it directly measures precipitation in the sky, updated every few minutes. However, IMD currently only makes rendered radar images public — not the underlying raw data needed to feed into an AI model. Because of this, VAYUNET treats radar honestly as a **future enhancement**, not a Phase 1 building block. If a formal research-data agreement with IMD becomes possible later, radar would meaningfully improve the system's short-term accuracy.

Ground observations

Automatic Weather Stations and rain gauges provide ground-truth readings that are useful for checking how accurate the system is. IMD's public portal for this data was restricted to the public around 2025, so VAYUNET should not assume live access to it, and should instead rely primarily on satellite and reanalysis data, using any ground data that is accessible mainly for validation rather than as a core input.

4. What Makes VAYUNET Different

AI-based weather nowcasting is not a new idea — large research teams at Google DeepMind and elsewhere have already built impressive systems like DGMR (a radar-based rainfall predictor) and MetNet (a satellite-based nowcasting model). VAYUNET does not claim to invent nowcasting itself. Its honest differentiation comes from the specific combination of choices below, each of which addresses a real gap in how existing systems work for India specifically:

- **Built for India, using India's own satellite:** Existing well-known systems like DGMR and MetNet were built and trained on UK or US radar and satellite data, which isn't available in the same form over India. VAYUNET is designed around INSAT and IMDAA data from the ground up.
- **Connects weather risk to actual ground impact:** Most nowcasting systems stop at “will it rain?” VAYUNET goes one step further by combining its weather prediction with terrain data to estimate “which specific valley is likely to flood,” which is far more directly useful for emergency response.
- **Explains its reasoning, not just its output:** Instead of only showing a colored risk map, VAYUNET aims to show which specific atmospheric signals (rising moisture, increasing instability) drove a given alert, in plain language — helping a non-expert official trust and understand the warning.
- **Designed to be lightweight enough to actually run:** Some of the most advanced global weather AI models (like GraphCast or Pangu-Weather) are enormous and built for medium-range global forecasting, not hyper-local short-term nowcasting — they're the wrong tool for this specific job. VAYUNET is deliberately scaled to be trainable and runnable with realistic student-team resources.

It's important to be honest about scope: VAYUNET's first working version will cover a limited region and season, not the entire country year-round. The path to national, all-season coverage is a real and valuable future goal, but the hackathon prototype should not overstate what has actually been built.

5. What VAYUNET Delivers, Its Limits, and What Comes Next

Expected output

- A map-based dashboard showing current severe-weather risk for a chosen region, updated as new data comes in.
- A short-term (2–6 hour) forecast timeline that a user can scroll through.
- Plain-language risk categories (Low, Moderate, High, Severe) rather than raw technical numbers.
- An explanation panel showing which atmospheric signals are driving a given alert.
- A flood-impact overlay showing which specific low-lying areas are most likely to be affected if heavy rain occurs.

Honest limitations

- The prototype works for a specific region and time period used for training — it will not automatically generalize to every part of India or every season without further training data.
- Radar data, which would meaningfully improve short-term accuracy, is not available for public/student use and is not part of the current version.
- The flood-routing calculation is a first approximation based on terrain shape — it is useful for prioritizing where to send help first, but it is not a substitute for a full, detailed hydrological survey.
- Rare, extreme events (the ones we care about most) are naturally underrepresented in historical data, which makes them inherently harder to predict with full confidence — this is a known, industry-wide challenge, not something unique to VAYUNET.

Future scope

- Formal data-access arrangements with IMD to incorporate raw radar data, which would substantially sharpen very-short-term (0–30 minute) predictions.
- Expansion from a single test region to full national coverage, trained across multiple seasons and terrain types.
- Direct integration with State Disaster Management Authority systems so that alerts reach officials and the public automatically, not just as a dashboard.
- Ongoing retraining as climate patterns shift over time, so the system doesn't gradually become outdated.

In short: VAYUNET is not trying to be the most complicated weather model ever built. It is trying to be the most useful, most honestly-scoped hyper-local severe-weather warning system a student team can realistically build — with a clear, credible path toward becoming a real operational tool for India.